

Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering

09/2023 – Present

- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi, Regional Organizational Team Member at ESTIEM; Student-Athlete (Sailing, Skiing & Windsurfing).

Kabataş Boys' High School

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00; Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Club.

Experience

Work and Travel Program Participant

06/2025 – 09/2025

- Gained international work experience in the USA, improving English communication, teamwork, and adaptability in a multicultural environment.

Educational Mentor at Derece Kampüsü

07/2023 – 09/2025

- Gave mentorship and guidance to students preparing for university entrance exam, helping them with academic and career development.

Technical Projects (@GitHub)

FinTech & Trading Systems — Python, Next.js, FastAPI, PostgreSQL, WebSockets, Docker

- Developed **AlphaTrigger** (a real-time trading simulator featuring a Z-Score mean reversion strategy and volatility-adjusted position sizing) and the **FinTech Intelligence Platform** (featuring a live limit-order matching engine, portfolio risk analyzer, and AI-powered insights).

Enterprise Admin Dashboard — Next.js 15+, Prisma, Tailwind CSS, Framer Motion

- Developed **Performo**, a real-time administrative dashboard with role-based access control, task management, automated user onboarding, and optimized database queries.

Systems & CPU Scheduling — x86-64 Assembly, GNU Toolchain, Linux Syscalls

- Implemented **SchedSim**, a CPU scheduling simulator written in x86-64 GNU Assembly using direct Linux syscalls to execute FCFS, SJF, SRTF, Priority, and Round Robin scheduling algorithms.

Interpreters & Language Design — C, Python, Lexer Design, Functional Paradigms

- Built **Spirelayer** and **Aura**, including a C-based command interpreter with custom backtracking lexer/parser and a Python functional-language interpreter with closures, higher-order functions, and lexical/dynamic scoping.

Algorithms & Matching Engines — Java, Graph Theory, Heaps, Complexity Analysis

- Developed **MatrixNet** (a network graph analysis tool implementing Tarjan's SCC algorithm and articulation point detection) and **GigMatch-Pro** (a candidate matching engine using max-heap data structures for priority-based job allocation).

Mobile Development, Game & WebAssembly Engineering — React Native, Expo, C++, Qt6, WebAssembly, Java

- Built **Smash Minesweeper** (a polished mobile app featuring a reducer-driven state engine, BFS flood-fill, and custom audio effects), **Qt 2048** (a C++/Qt6 clone cross-compiled to WebAssembly with undo history), and **Nightpass** (a terminal-based survival card game with a state-machine rule engine).

Skills

- **Programming:** Python, Java, C/C++, Qt6, WebAssembly, TypeScript, JavaScript, SQL, x86-64 Assembly, Bash
- **Low-Level & Systems:** Linux Syscalls, Process Scheduling, Multithreading (Pthreads), Memory Management, Qt Framework
- **Enterprise Full-Stack & Mobile & AI:** Next.js 15+, React Native, Expo, React, Prisma, SQLAlchemy, FastAPI, Tailwind CSS, PostgreSQL, AI-Assisted Development
- **Tools & Ecosystem:** Linux/Unix, Git/GitHub, Docker, Alembic, ccache, Pytest, CI/CD, Makefile, LaTeX, MS Office
- **Languages:** Turkish (native), English (fluent), German (elementary)

Volunteer Experience

- **TOÇEV:** Provided volunteer tutoring and mentoring for children from disadvantaged backgrounds. 09/2019 – 03/2020
- **Boğaziçi University Industrial Engineering Club:** Actively contributed to the Volunteering Division. 09/2023 – 06/2024
- **Kabataş Boys' High School CIP Club:** Organized and executed various community-oriented projects. 09/2018 – 06/2023